

Extending the PREVENT Equations with Cardiac MRI: Prediction of 10-year Heart Failure Risk

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See also the editorial by Hanneman and Moayed in this issue.

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Background: Accurate prediction of heart failure (HF) risk is crucial for early intervention and prevention. Traditional models rely on clinical risk factors, whereas cardiac MRI parameters may offer additional predictive value.

Purpose: To evaluate whether integrating multidimensional cardiac MRI parameters with the Predicting Risk of Cardiovascular Disease EVENTS (PREVENT) equations improves HF risk prediction.

Materials and Methods: This secondary analysis of a prospective study included participants from the UK Biobank who underwent cardiac MRI. Participants were randomly divided into 70% for a training set and 30% for an internal test set. Eighty-two representative cardiac MRI variables were log-transformed and standardized. Variable selection was performed using least absolute shrinkage and selection operator (LASSO) regression combined with bootstrap resampling. Sex-specific Fine-Gray competing risk models were constructed based on the PREVENT equations. Model performance was evaluated using the C-index and calibration charts.

Results: This study included 39 069 participants (mean age, 55 years $\pm$ 7.5 [SD]; 20 520 women). Sixteen cardiac MRI parameters identified by LASSO enhanced HF risk prediction. In the training set ($n = 27\,531$), the C-index increased from 0.753 (95% CI: 0.728, 0.777) to 0.812 (95% CI: 0.787, 0.837) (Δ C-index = 0.059; $P < .001$); in the internal test set ($n = 11\,538$), it increased from 0.760 (95% CI: 0.719, 0.801) to 0.821 (95% CI: 0.779, 0.862) (Δ C-index = 0.061; $P = .007$). In men, left atrial (LA) functional parameters contributed most to the improvement in C-index, including LA ejection fraction (Δ C-index = 0.022), whereas in women, the gain was primarily driven by left ventricular (LV) ejection fraction (Δ C-index = 0.011) and myocardial strain (LV circumferential strain: Δ C-index = 0.008; LV radial strain: Δ C-index = 0.006).

Conclusion: Integrating cardiac MRI parameters into the PREVENT equations improved HF risk prediction and highlighted sex-specific pathophysiologic differences.

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Cardiology has made immense progress over the past few decades, yet heart failure (HF) remains a growing global epidemic (1). It is a leading cause of cardiovascular disease morbidity and mortality, with associated health care costs rising annually. This places a heavy burden on society and the economy, highlighting the critical need for precise risk stratification and intervention in the early stages of the disease (2).

In 2023, the American Heart Association introduced the Predicting Risk of Cardiovascular Disease EVENTS (PREVENT) equations, a race-free model incorporating cardiovascular, kidney, and metabolic factors to estimate 10-year risks of atherosclerotic cardiovascular disease and HF (3). Validated in 6.6 million U.S. adults, the model demonstrated excellent discrimination and calibration. Compared with traditional tools such as the Framingham Risk Score and the QRISK cardiovascular disease risk algorithm, PREVENT offers broader outcome coverage and specific HF prediction (3,4). However, current PREVENT equations rely primarily on conventional clinical risk factors. Given that the onset and progression of HF are closely associated with myocardial structural changes, ventricular functional abnormalities, and atrial remodeling detectable at imaging (5), existing models may overlook subclinical imaging biomarkers (6). There remains an urgent need to improve

predictive accuracy to identify individuals at increased susceptibility to future HF events (6).

Cardiac MRI, as a noninvasive, accurate, and reproducible tool, enables the precise quantification of ventricular volumes, mass, myocardial fibrosis, and strain parameters (7) and has received a Class I recommendation in the 2021 European Society of Cardiology HF guidelines (8). Existing studies demonstrate that cardiac MRI parameters provide independent and incremental predictive value for adverse outcomes such as HF, even after adjustment for conventional risk factors (7,9). However, most evidence is derived from specific high-risk populations with limited sample sizes and has focused on single cardiac MRI parameters, with a lack of integration and validation in general populations for long-term HF risk prediction (10,11).

This study aims to evaluate whether integrating multidimensional cardiac MRI parameters with the PREVENT equations improves 10-year HF risk prediction.

Materials and Methods

Study Sample

This secondary analysis of a prospective study included participants from the UK Biobank, a large prospective cohort for

Abbreviations

HF = heart failure, LA = left atrial, LASSO = least absolute shrinkage and selection operator, LV = left ventricular, LVEF = LV ejection fraction, PREVENT = Predicting Risk of Cardiovascular Disease EVENTS

Summary

Integrating cardiac MRI parameters into the Predicting Risk of Cardiovascular Disease EVENTS equations improved the prediction of incident heart failure risk and uncovered sex-specific pathophysiologic differences.

Key Results

- In this secondary analysis of a prospective study of 39 069 participants, integration of 16 cardiac MRI parameters with the Predicting Risk of Cardiovascular Disease EVENTS equations improved heart failure (HF) risk prediction.
- The C-index increased from 0.753 to 0.812 ($P < .001$) in the training set and from 0.760 to 0.821 ($P = .007$) in the internal test set.
- Left atrial ejection fraction contributed most to HF risk prediction in men (Δ C-index = 0.022), whereas left ventricular ejection fraction (Δ C-index = 0.011) drove the improvement in women.

the investigation of biologic and environmental determinants of health. Institutional review board approval for the UK Biobank study was obtained from the North West Multi-centre Research Ethics Committee (no. 21/NW/0157), and all participants provided written informed consent. The current study was conducted using the UK Biobank resource (project no. 90492). The study sample included participants recruited between March 2006 and December 2022. Baseline assessments included questionnaires, interviews, physical measurements, and blood sampling (12). Participants with at least two available cardiac MRI parameters were included. Those with prevalent HF, coronary heart disease, stroke, or missing diagnostic information or those who were lost to follow-up before cardiac MRI were excluded (Fig 1).

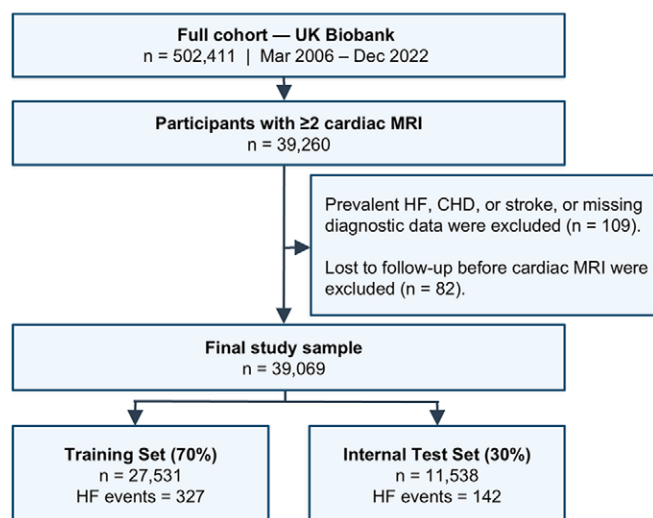


Figure 1: Flow diagram of participant selection. CHD = coronary heart disease, HF = heart failure.

Cardiac MRI Procedures

The UK Biobank imaging study aimed to scan 20% of the cohort using multiorgan MRI, with the protocol described in detail elsewhere (13). Cardiac MRI examinations were performed using standardized 1.5-T scanners (MAGNETOM Aera, Siemens) without contrast or stress agents. The protocol included anatomic and cine imaging, myocardial tagging, T1 mapping, and aortic flow, analyzed with cvi42 software (Circle Cardiovascular Imaging) to derive ventricular, atrial, and aortic parameters (13,14). All cardiac MRI metrics used in this study were image-derived phenotypes obtained from the UK Biobank centralized analysis pipeline.

Cardiac MRI Analysis

Cardiac MRI parameters were obtained directly from the UK Biobank centrally processed imaging dataset, as previously described (15). All parameters were derived using an automated machine learning–based analysis pipeline; details of image segmentation, quality control procedures, and code availability have been reported in that publication (https://github.com/baiwenjial/ukbb_cardiac). As this study used preprocessed deidentified imaging parameters rather than primary image analysis, independent reader assessment and blinding procedures were not applicable.

Outcome Definition and Covariates

Incident HF was defined as the first recorded occurrence of *International Classification of Diseases, Tenth Revision*, codes I50.0, I50.1, or I50.9 from self-reports or primary care, hospital inpatient, and death registry data (Table S1). Core covariates for the PREVENT equations included age, sex, systolic blood pressure, body mass index (calculated as weight in kilograms divided by height in meters squared), smoking status, diabetes mellitus, anti-hypertensive medication use, and estimated glomerular filtration rate (3). Additional analyses were adjusted for demographic characteristics, lifestyle, lipids, clinical history, and other medications (see Table S2 for detailed definitions).

Cardiac MRI Metrics Selection and Model Derivation

In the training set, a total of 82 cardiac MRI variables (listed in Table S3) were log-transformed to approximate a normal distribution and standardized to units of SD. To avoid scaling bias, uncorrected metrics were used in place of body surface area–indexed values (16), with body mass index included as a covariate to allow flexible adjustment for body size. To reduce collinearity and improve clinical applicability, candidate predictors were restricted to 28 global cardiac MRI indexes (17). On the basis of Pearson correlation analysis, variable pairs with absolute correlation over 0.80 were filtered using an iterative mean absolute correlation criterion, retaining 19 nonredundant variables for subsequent selection (Table S4).

Least absolute shrinkage and selection operator (LASSO) regression was then applied to the selected variables, with the optimal λ parameter determined by means of 10-fold cross-validation based on minimum validation error. To ensure selection stability, 1000 bootstrap repetitions of the LASSO procedure were performed; cardiac MRI variables selected in at least 95% of repetitions were defined as final candidates. This threshold was chosen based on prior studies aimed at improving model generalizability

and reducing overfitting (18). Models were first constructed using the selected cardiac MRI variables alone to assess their independent predictive value for HF incidence. Building on this, accounting for death as a competing risk, sex-specific prediction models were developed in the training set with use of Fine-Gray competing risk regression. Additional methodologic details, including assessment of the proportional hazards assumption, evaluation of nonlinearity using restricted cubic splines, sensitivity analysis using body surface area-indexed cardiac MRI parameters, heuristic shrinkage factor calculation, and the full 10-year HF risk prediction formula, are provided in Appendix S1, Tables S5 and S6, and Figure S1.

Evaluation of predictive model performance

To evaluate model discrimination, the Harrell C-index adjusted for competing risks was calculated using the R package riskRegression (R Foundation). Calibration was assessed by comparing mean predicted risk and observed risk across deciles of predicted risk with use of the Greenwood-Nam-D'Agostino goodness-of-fit test.

Associations of selected cardiac MRI metrics with HF

In the internal test set, each cardiac MRI parameter was individually assessed for its association with 10-year HF risk by means of Cox proportional hazards regression. Hazard ratios (HRs) and 95% CIs were calculated separately for male and female participants, adjusted for multiple covariates. The Benjamini-Hochberg procedure was applied to control the false discovery rate for multiple testing. Each cardiac MRI variable was entered individually into the model to evaluate its incremental contribution to the C-index of the model (Fig S2). Subsequently, a stepwise forward analysis was performed where variables were added sequentially in descending order of LASSO importance. At each step, the incremental gain in C-index (Δ C-index) and 95% CI were calculated relative to the preceding model to quantify the significance of the added information.

Sensitivity Analyses

Five sensitivity analyses were performed. First, analyses were repeated in participants with complete data for all 28 global cardiac MRI variables without imputation, using bootstrap LASSO with 1000 resamples to identify the most relevant predictors. Second, alternative selection algorithms—including random forest feature importance ranking and stepwise regression—were applied and compared with the primary LASSO results to evaluate consistency in predictor selection. Third, bootstrap LASSO was applied to a comprehensive candidate set including global and segmental cardiac MRI variables to assess whether regional functional metrics could further enhance predictive performance. Fourth, the PREVENT equations were recalibrated to address potential differences in baseline risk between the training set and the UK Biobank population. Finally, to benchmark the performance of the new model, its predictive accuracy was compared against existing HF risk models, including the Pooled Cohort Equations to Prevent Heart Failure and the Atherosclerosis Risk in Communities HF risk score. This assessment was restricted to a subset of participants who possessed complete data for all specific variables required to calculate these external risk scores.

Data Availability

The data used for analysis in this study are available from the UK Biobank project site, subject to registration and an application process. Further details can be found at <https://www.ukbiobank.ac.uk>. Two authors (W.X. and F.Z.) had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Statistical Analysis

Statistical analyses were performed by two authors (M.Z. and Y.D.). Two-sided $P < .05$ was considered indicative of statistically significant difference. Missing data (ranging from 0% to 14.7%) were imputed using a random forest algorithm, with comparisons of preimputation and postimputation distributions detailed in Tables S7 and S8. The dataset was randomly split into a training set (70%) and an internal test set (30%). Baseline characteristics were compared using analysis of variance or the χ^2 test. The software used for this study included R (version 4.3.0) and SAS (version 9.4; SAS Institute).

Results

Participant Characteristics

Of 39 260 participants with at least two cardiac MRI measurements in the UK Biobank imaging study, 191 were excluded due to prevalent HF or missing diagnostic information ($n = 65$), prevalent coronary heart disease or stroke ($n = 44$), or loss to follow-up before cardiac MRI ($n = 82$), leaving a final study sample of 39 069 participants (mean age, 55 years $\pm$ 7.5 [SD]; age range, 40–71 years; 20 520 women [52.5%]) (Fig 1, Table 1). Participants were randomly assigned to a training set ($n = 27 531$, 327 incident HF events) and an internal test set ($n = 11 538$, 142 incident HF events), with comparable baseline characteristics across sets (all $P > .05$) (Table 1). Sex-specific differences in baseline characteristics are presented in Table S9. Men had higher body mass index, blood pressure, smoking and alcohol consumption rates, and a higher prevalence of hypertension and diabetes (all $P < .001$), whereas women exhibited higher high-density lipoprotein cholesterol levels ($P < .001$).

Volumetric indexes within identical anatomic structures exhibited high collinearity ($|r| > 0.80$), necessitating careful variable selection to address collinearity during model development in the training set (Fig S3). In contrast, myocardial strain parameters showed only weak-to-moderate correlations with volumetric and ejection fraction indexes (eg, end-diastolic volume, end-systolic volume, and stroke volume, $|r| < 0.50$), indicating that myocardial strain provides complementary functional information distinct from traditional volumetric measures.

Associations of Cardiac MRI Metrics with HF

With use of LASSO regression, 16 cardiac MRI parameters were selected from 28 candidates for their association with HF risk (Fig 2). These variables encompassed ventricular systolic function, chamber volume and load, atrial function, and large-artery elasticity. Most indexes showed consistent associations in both sexes (eg, left ventricular [LV] ejection fraction [LVEF]: male HR, 0.67 [95% CI: 0.62, 0.72], $P < .001$; female HR, 0.57 [95% CI: 0.49, 0.66], $P < .001$). However, ascending aorta

Table 1: Baseline Characteristics of Participants

Characteristic	All Participants (N = 39 069)	Training Set (n = 27 531)	Internal Test Set (n = 11 538)	P Value*
Age (y) [†]	55 ± 7.5 (40–71)	55 ± 7.5 (40–71)	55.3 ± 7.5 (40–71)	.51
Race				.17
White	37 823 (96.81)	26 675 (96.89)	11 148 (96.62)	
Other races [‡]	1246 (3.19)	856 (3.11)	390 (3.38)	
Higher education [§]	24 024 (61.49)	16 957 (61.59)	7067 (61.25)	.53
Body mass index	26.56 ± 4.22	26.56 ± 4.22	26.56 ± 4.23	.96
Exercise	31 181 (79.81)	21 992 (79.88)	9189 (79.64)	.60
Current smoking	2432 (6.22)	1713 (6.22)	719 (6.23)	.99
Current alcohol consumption	29 768 (76.19)	20 972 (76.18)	8796 (76.24)	.91
SBP (mm Hg)	135.13 ± 17.70	135.16 ± 17.67	135.04 ± 17.78	.55
DBP (mm Hg)	81.51 ± 9.89	81.51 ± 9.85	81.50 ± 9.98	.90
eGFR (mL/min/1.73 m ²)	95.14 ± 11.87	95.15 ± 11.85	95.10 ± 11.92	.71
Total cholesterol (mmol/L)	5.74 ± 1.08	5.74 ± 1.08	5.74 ± 1.08	.88
LDL cholesterol (mmol/L)	3.59 ± 0.83	3.59 ± 0.83	3.59 ± 0.83	.84
HDL cholesterol (mmol/L)	1.48 ± 0.37	1.48 ± 0.38	1.48 ± 0.37	.73
Clinical diagnosis				
Hypertension	17 894 (45.80)	12 606 (45.79)	5288 (45.83)	.95
Diabetes	1204 (3.08)	853 (3.10)	351 (3.04)	.79
Medication use				
Antihypertensive drugs	4983 (12.75)	3529 (12.82)	1454 (12.60)	.57
Antihyperglycemic drugs	672 (1.72)	467 (1.70)	205 (1.78)	.61
Lipid-lowering drugs	4159 (10.65)	2973 (10.80)	1186 (10.28)	.13

Note.—Data are numbers of participants (with percentages in parentheses) for categorical variables or means ± SDs for continuous variables. DBP = diastolic blood pressure, eGFR = estimated glomerular filtration rate, HDL = high-density lipoprotein, LDL = low-density lipoprotein, SBP = systolic blood pressure.

* P values represent differences between the training and internal test sets. Continuous variables were compared using the independent samples *t* test. Binary categorical variables were compared using the χ^2 test.

[†] Data in parentheses are ranges.

[‡] Other races include Asian, Black, mixed, and other self-reported categories.

[§] Higher education is defined as self-reported attainment of a college or university degree or other professional qualifications.

^{||} Body mass index was calculated as patient weight in kilograms divided by patient height in meters squared.

distensibility and right ventricular stroke volume showed associations in men (HR, 1.32 [95% CI: 1.03, 1.69], *P* = .025; HR, 0.76 [95% CI: 0.63, 0.92], *P* = .004, respectively) but not in women (*P* = .758 and .630, respectively), whereas right atrial stroke volume was not statistically significant in either sex (male *P* = .404; female *P* = .288).

HF Risk Prediction with Cardiac MRI Metrics

In the model using only the 16 cardiac MRI variables, the C-index was 0.791 (95% CI: 0.764, 0.817) in the training set and 0.801 (95% CI: 0.759, 0.842) in the internal test set (Table S10). Table 2 presents the main predictive performance metrics before and after incorporating the 16 cardiac MRI variables into the PREVENT equations. In the training set, the overall C-index increased from 0.753 (95% CI: 0.728, 0.777) to 0.812 (95% CI: 0.787, 0.837) (Δ C-index = 0.059; *P* < .001). In the internal test set, the overall C-index rose from 0.760 (95% CI: 0.719, 0.801) to 0.821 (95% CI: 0.779, 0.862) (Δ C-index = 0.061; *P* = .007), with greater gains in men (Δ C-index = 0.086; *P* < .001) than women (Δ C-index = 0.043; *P* = .035).

The cardiac MRI-integrated model reduced calibration deviations observed in the base PREVENT equations, as supported by the Greenwood-Nam-D'Agostino goodness-of-fit test (male: χ^2 = 1.09, *P* = .87; female: χ^2 = 3.70, *P* = .93) versus

the base equations (male: χ^2 = 9.90, *P* = .21; female: χ^2 = 7.69, *P* = .46), supporting enhanced clinical applicability (Fig 3). Sex-specific β -coefficients and baseline survival estimates of the model combining PREVENT equations and cardiac MRI metrics are provided in Table 3.

Figure 4 illustrates the incremental improvement in C-index as the 16 cardiac MRI variables were sequentially added in the internal test set. In men, left atrial (LA) ejection fraction (Δ C-index = 0.022 [95% CI: 0.015, 0.028]) and LA minimum volume (Δ C-index = 0.015 [95% CI: 0.009, 0.021]) contributed most, whereas in women, LVEF (Δ C-index = 0.011 [95% CI: 0.006, 0.015]) and LV circumferential strain (Δ C-index = 0.008 [95% CI: 0.005, 0.015]) provided the largest incremental gains.

Sensitivity Analysis

In the sensitivity analysis without data imputation, bootstrap LASSO with 1000 resamples identified 16 cardiac MRI variables consistent with the primary analysis, including LVEF, LV circumferential strain, LA ejection fraction, and LA minimum volume, among others (Table S11). Second, methodologic stability was supported by alternative feature selection approaches. Random forest feature importance (top 16) showed substantial overlap with the primary LASSO model (*P* = .61 vs the primary model); stepwise regression identified a parsimonious set of 10 predictors

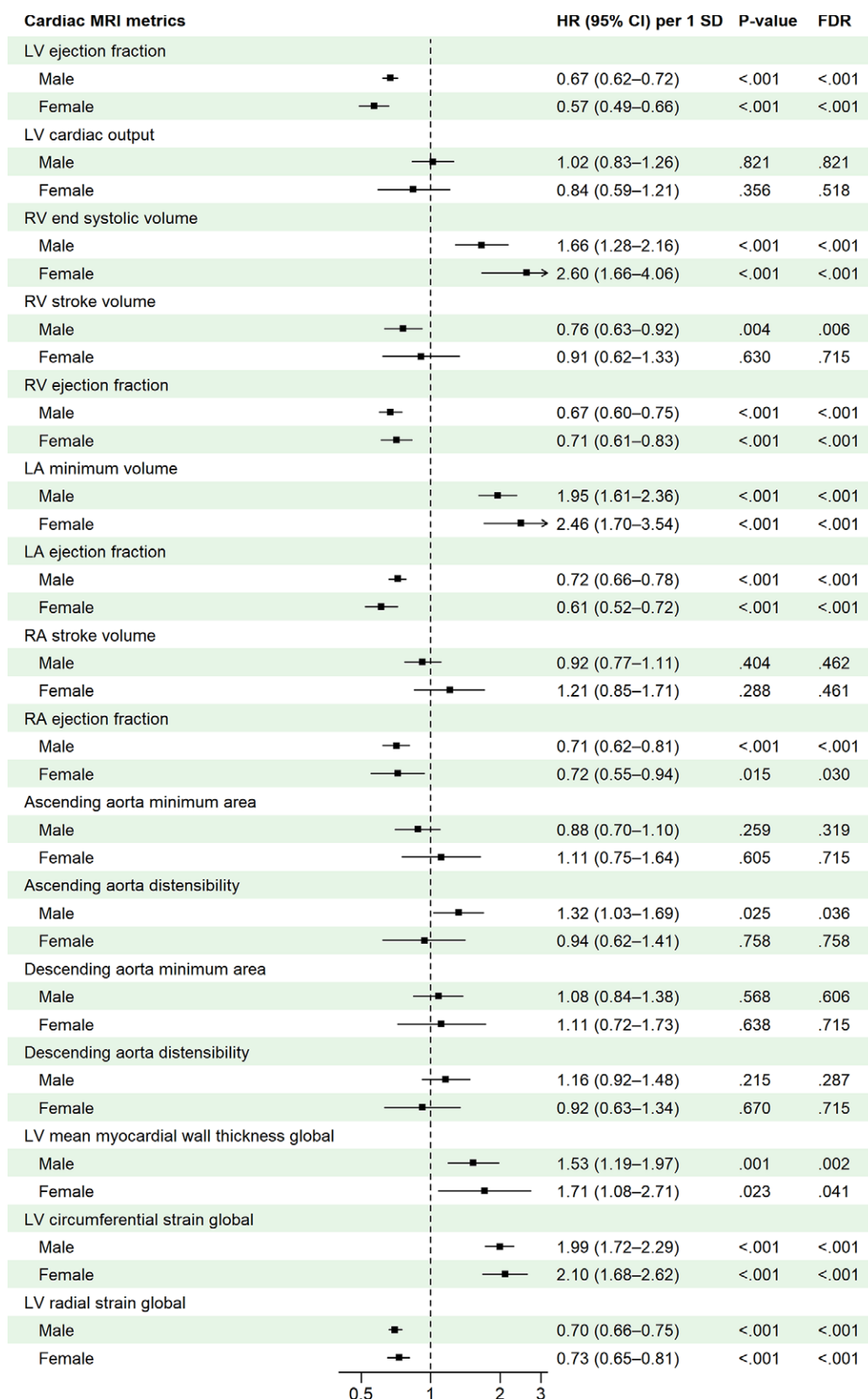


Figure 2: Forest plot shows associations between cardiac MRI parameters and incident heart failure by sex in the internal test set ($n = 11\,538$). Cox proportional hazards regression was used to estimate hazard ratios (HRs) per 1-SD increase in each log-transformed cardiac MRI parameter, adjusted for age, sex, race, body mass index, current smoking, current alcohol consumption, exercise, hypertension, diabetes, low-density lipoprotein cholesterol, antihypertensive drugs, antihyperglycemic drugs, and lipid-lowering drugs. P values were obtained from Cox regression and corrected for multiple testing with use of the Benjamini-Hochberg false discovery rate (FDR) method. The dashed vertical line indicates an HR value of 1. Data points represent HRs; error bars represent 95% CIs. Left ventricular (LV) ejection fraction was the strongest protective predictor in both sexes, whereas left atrial (LA) minimum volume showed a greater risk-increasing effect in women than in men. RA = right atrial, RV = right ventricular.

Table 2: Predictive Performance of the PREVENT Equations for 10-year HF Risk Before and After Extension with Cardiac MRI Parameters

Metric	Overall	Men	Women
70% training set (<i>n</i> = 27 531)			
No. of participants with HF*	327/27 531 (1.19)	218/13 077 (1.67)	109/14 454 (0.75)
C-index, PREVENT	0.753 (0.728, 0.777)	0.732 (0.700, 0.764)	0.741 (0.701, 0.782)
C-index, PREVENT + cardiac MRI	0.812 (0.787, 0.837)	0.810 (0.780, 0.841)	0.801 (0.757, 0.845)
Δ C-index [†]	0.059	0.078	0.060
<i>P</i> value [‡]	<.001	<.001	.007
30% internal test set (<i>n</i> = 11 538)			
No. of participants with HF*	142/11 538 (1.23)	105/5472 (1.92)	37/6066 (0.61)
C-index, PREVENT	0.760 (0.719, 0.801)	0.715 (0.661, 0.769)	0.774 (0.701, 0.848)
C-index, PREVENT + cardiac MRI	0.821 (0.779, 0.862)	0.801 (0.753, 0.850)	0.817 (0.733, 0.901)
Δ C-index [†]	0.061	0.086	0.043
<i>P</i> value [‡]	.007	<.001	.035

Note.—C-index is presented with 95% CIs in parentheses. Predicting Risk of Cardiovascular Disease EVENTS (PREVENT) represents the base model including age, sex, systolic blood pressure, body mass index, smoking status, diabetes mellitus, antihypertensive medication use, and estimated glomerular filtration rate. PREVENT + cardiac MRI represents the PREVENT base model extended with 16 selected cardiac MRI parameters: left ventricular (LV) ejection fraction, LV cardiac output, LV mean wall thickness, LV circumferential strain, LV radial strain, right ventricular end-systolic volume, right ventricular stroke volume, right ventricular ejection fraction, left atrial minimum volume, left atrial ejection fraction, right atrial stroke volume, right atrial ejection fraction, ascending aorta minimum area, ascending aorta distensibility, descending aorta minimum area, and descending aorta distensibility. HF = heart failure.

* Data in parentheses are percentages.

[†] Δ C-index represents the difference in C-index between the PREVENT + cardiac MRI model and the PREVENT base model.

[‡] *P* values were obtained using the DeLong method for comparison of correlated C-indexes.

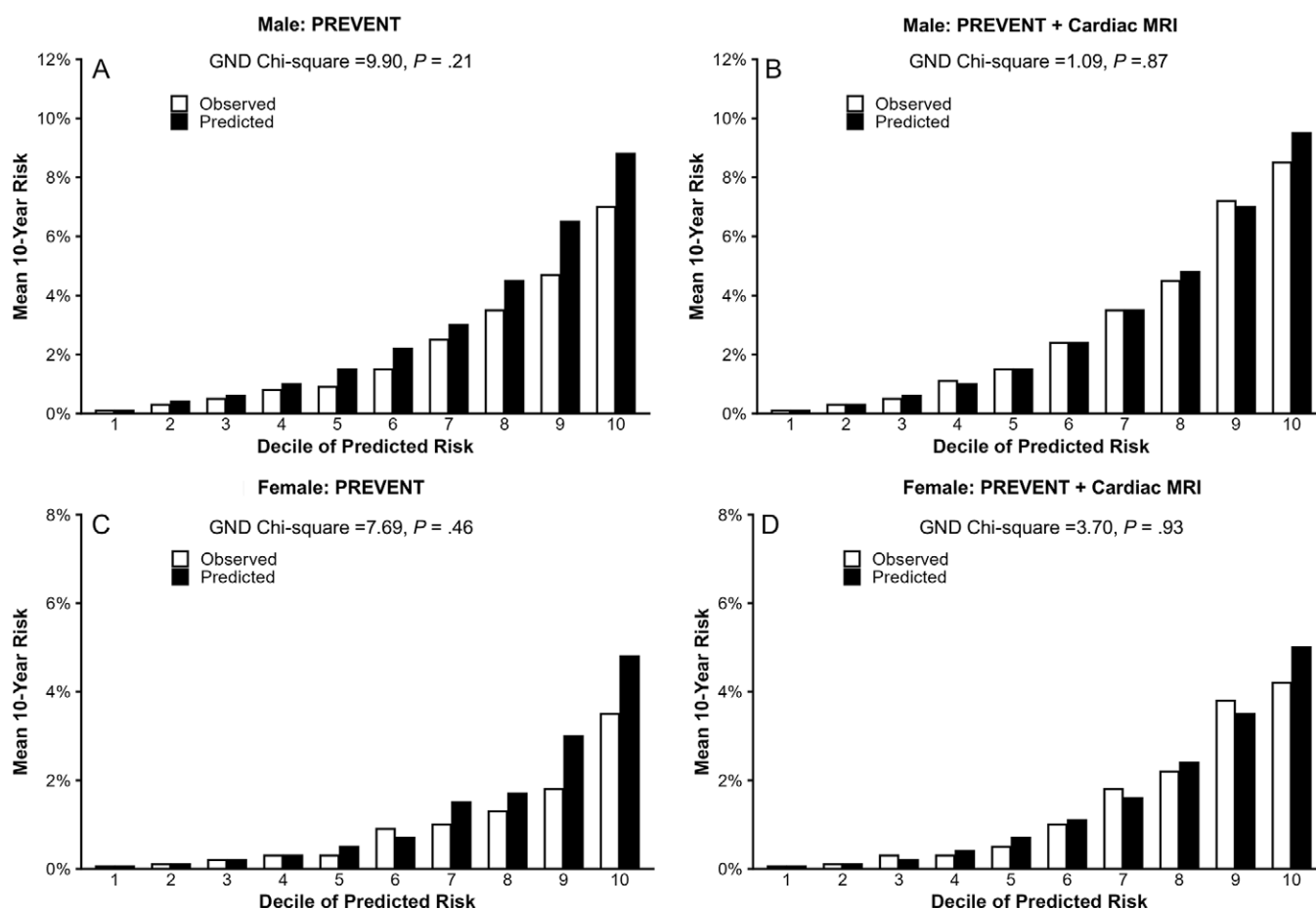


Figure 3: Calibration bar charts compare predicted and observed 10-year heart failure risk for the Predicting Risk of Cardiovascular Disease EVENTS (PREVENT) base model and the cardiac MRI-extended model in the internal test set (*n* = 11 538). Panels (A) and (C) show the calibration of the base PREVENT equations in men and women, respectively. Panels (B) and (D) display the calibration of the extended model integrating 16 cardiac MRI parameters. The goodness-of-fit test was evaluated using the Greenwood-Nam-D'Agostino (GND) test; χ^2 statistics and *P* values are provided above each plot. The PREVENT and cardiac MRI model demonstrated improved calibration compared with the base model in both men (GND χ^2 = 1.09; *P* = .87 vs χ^2 = 9.90; *P* = .21) and women (GND χ^2 = 3.70; *P* = .93 vs χ^2 = 7.69; *P* = .46).

Table 3: β -Coefficients of the PREVENT Equations Extended by 16 Cardiac MRI Metrics for 10-year Prediction of Heart Failure Events in the Training Set

Variable	β -Coefficient	
	Men	Women
Baseline survival $S_0(10)^*$	0.9581	0.9152
Base risk estimator, logit (PREVENT risk) [†]	0.8965	0.9208
Cardiac MRI parameters [‡]		
LV ejection fraction	-0.1657	-0.4654
LV cardiac output	0.1610	0.3984
RV end-systolic volume	-0.1168	0.1355
RV stroke volume	-0.0277	-0.1013
RV ejection fraction	-0.0274	0.2249
LA minimum volume	0.2378	0.1269
LA ejection fraction	-0.0332	-0.1672
RA stroke volume	-0.0183	-0.0996
RA ejection fraction	-0.0738	-0.0748
Ascending aorta minimum area	0.0205	0.0492
Ascending aorta distensibility	0.0587	-0.0924
Descending aorta minimum area	0.2100	0.0549
Descending aorta distensibility	0.0930	0.1740
LV mean wall thickness	0.2121	0.1766
LV circumferential strain	0.5311	0.2282
LV radial strain	0.2209	0.0748

Note.—LA = left atrial, LV = left ventricular, PREVENT = Predicting Risk of Cardiovascular Disease EVENTS, RA = right atrial, RV = right ventricular.

* Baseline survival probability at 10 years in the Fine-Gray competing risk model.

[†] Linear predictor calculated using the original PREVENT equations for heart failure.

[‡] Cardiac MRI variables were log-transformed and standardized (Z-scores) based on the mean and SD of the training set before model fitting.

with comparable performance ($P = .29$ vs the primary model). Key functional parameters—including LVEF, LV circumferential strain, LA ejection fraction, and ascending aorta distensibility—were consistently retained across methods. Discriminative performance of the alternative models remained comparable to the primary model in the internal test set (stepwise regression: C-index = 0.825 [95% CI: 0.786, 0.864]; random forest top 10: C-index = 0.825 [95% CI: 0.786, 0.865]) (Fig S4, Table S12).

The sensitivity analysis incorporating segmental cardiac MRI metrics identified 20 predictors—the 16 primary variables plus LV longitudinal strain global, LV circumferential strain in American Heart Association segment 11, LV radial strain in segment 11, and LV radial strain in segment 4 (Table S13). However, this expanded model showed no evidence of improved predictive performance over the primary model in the internal test set (C-index: 0.823 [95% CI: 0.781, 0.864] vs 0.821 [95% CI: 0.779, 0.862], $P = .53$), supporting retention of the parsimonious global-variable model.

Recalibration of the PREVENT equations did not alter the discriminative performance of the base model (men: C-index = 0.715 [95% CI: 0.661, 0.770]; women: 0.774 [95% CI: 0.701, 0.848]), and cardiac MRI augmentation continued to improve accuracy after recalibration (men: 0.796 [95% CI: 0.732, 0.859];

women: 0.824 [95% CI: 0.726, 0.918]) (Table S14). Notably, the cardiac MRI-extended model demonstrated superior discrimination in women, achieving a C-index of 0.861 (95% CI: 0.815, 0.900) and surpassing both the Atherosclerosis Risk in Communities score (0.804 [95% CI: 0.718, 0.890], $P < .001$) and the Pooled Cohort Equations to Prevent Heart Failure (0.793 [95% CI: 0.688, 0.899], $P = .005$), highlighting its enhanced value for female-specific risk stratification (Table 4).

Discussion

To our knowledge, this study is the first to systematically evaluate the incremental value of multidimensional cardiac MRI parameters for the Predicting Risk of Cardiovascular Disease EVENTS model in predicting 10-year heart failure (HF) risk within a large prospective cohort. Incorporating 16 key cardiac MRI variables improved risk prediction, with the C-index in the internal test set increasing from 0.760 to 0.821 (Δ C-index = 0.061, $P = .007$). Notably, predictive gains in men were primarily driven by atrial measures (eg, left atrial ejection fraction), whereas in women, ventricular parameters (eg, left ventricular ejection fraction and myocardial strain) contributed most. These findings provide imaging-based evidence for individualized HF risk prediction.

Previous reviews have noted that most HF risk prediction models achieve moderate discrimination (C-index >0.70) but rely largely on clinical variables and may be affected by overfitting or missing data, limiting clinical applicability (19). Traditional predictors, including age, LVEF, and B-type natriuretic peptide, remain central, whereas emerging biomarkers, particularly cardiac MRI parameters such as myocardial strain and LA structure, improve predictive accuracy (19,20). In patients with myocardial infarction, global longitudinal strain measured at cardiac MRI outperformed LVEF in the prediction of major adverse cardiovascular events (area under the receiver operating characteristic curve: 0.73 vs 0.64) (17). Adding global longitudinal strain to clinical risk factors increased the C-index to 0.81, exceeding models using LV end-diastolic volume and LVEF (0.77), highlighting incremental value (21). Raisi-Estabragh and McCracken et al (20) reported that LA structure and function independently predict cardiovascular incidence and mortality beyond LV metrics. Research by Gutman et al (22) emphasized that HF model performance depends more on biologically relevant covariate diversity than algorithm choice, underscoring the value of incorporating cardiac MRI indicators. Although prior studies focused on single cardiac MRI variables, our study integrated multidimensional cardiac MRI metrics in a large prospective cohort, enhancing HF risk prediction and enabling precise identification of high-risk individuals.

Cardiac MRI serves as the reference standard for evaluating cardiac structure and function, providing biventricular metrics such as LVEF and right ventricular ejection fraction (23). In this study, ventricular parameters in women contributed most to C-index improvement for HF prediction, consistent with prior reports (24), whereas atrial parameters added the greatest incremental value in men. Although LA ejection fraction has been shown to predict HF onset two- to threefold better than LV indexes, sex-specific differences have not been previously addressed (20). Our results suggest that HF pathophysiology may involve earlier atrial remodeling in men and earlier ventricular impairment in women, reflecting distinct myocardial remodeling, inflammation, fibrosis, and hormonal regulation patterns.

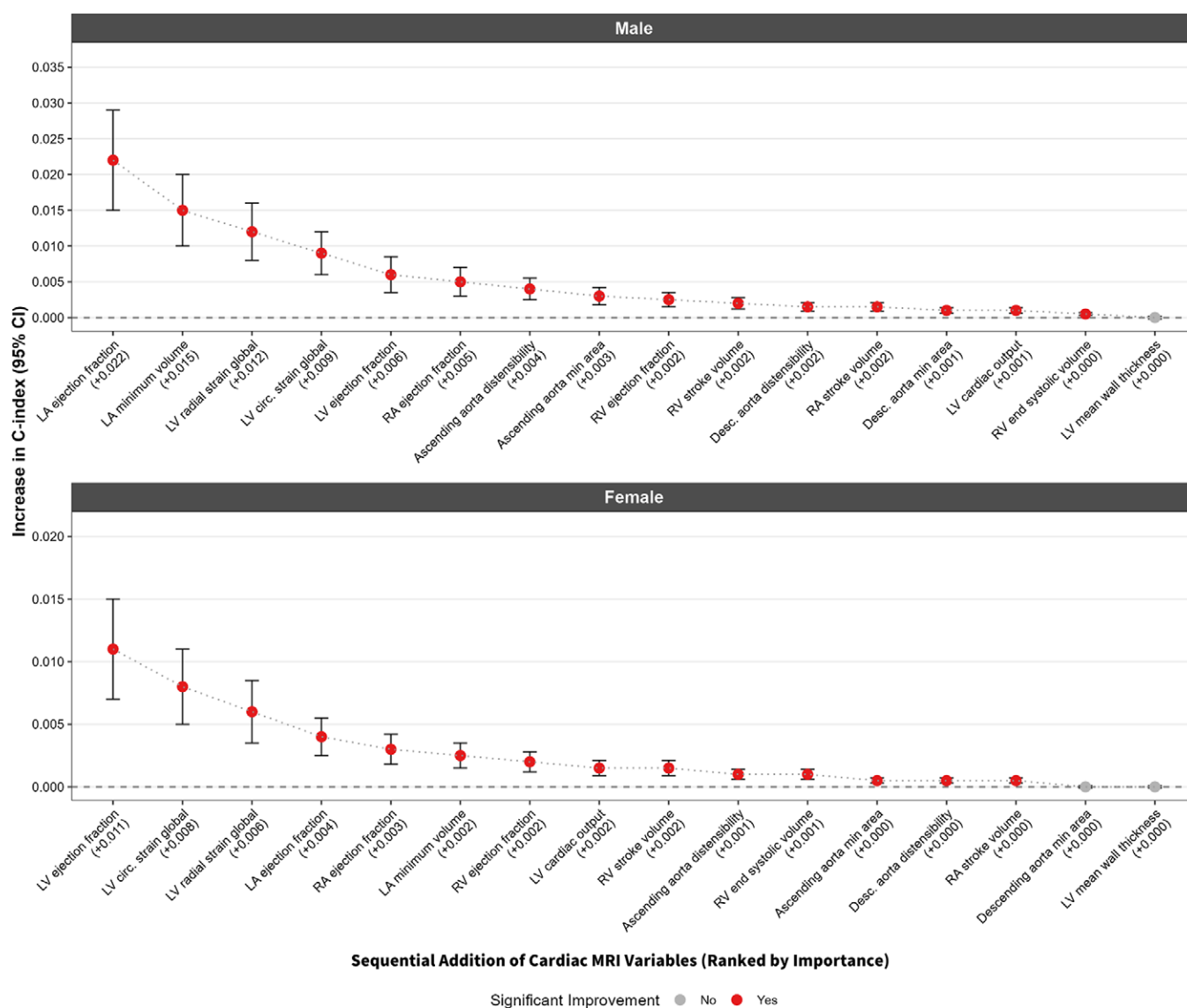


Figure 4: Dot plots show stepwise incremental improvement in C-index upon sequential addition of cardiac MRI variables to the Predicting Risk of Cardiovascular Disease EVENTS (PREVENT) equations by sex in the internal test set ($n = 11\,538$). The base model represents the PREVENT equations. Standardized cardiac MRI variables were added sequentially based on their marginal contribution to predictive accuracy, defined by their selection frequency during bootstrap least absolute shrinkage and selection operator variable selection. Each data point represents the mean incremental gain in C-index (Δ C-index) at each step relative to the preceding model, with error bars indicating 95% CIs derived from 1 000 bootstrap resamplings. Red dots indicate a statistically significant improvement (lower 95% CI bound > 0); gray dots indicate nonsignificant improvement. The dashed line connecting data points illustrates the cumulative trend of incremental C-index improvement across sequentially added variables. Data for men are presented in the upper panel, and data for women in the lower panel. In men, left atrial (LA) ejection fraction contributed the greatest incremental improvement (Δ C-index = 0.022), whereas in women, left ventricular (LV) ejection fraction contributed most (Δ C-index = 0.011). Circ. = circumferential, Desc. = descending, min = minimum, RA = right atrial, RV = right ventricular.

Under pressure overload, men's cardiomyocytes tend to elongate, leading to eccentric hypertrophy with secondary LA enlargement, whereas women's cardiomyocytes thicken, causing concentric remodeling with preserved ejection fraction but diastolic dysfunction (25). Women exhibit higher levels of inflammatory mediators, more active pathways, and increased autoimmune prevalence, promoting fibrosis and stiffness detectable by myocardial strain (25). Postmenopausal estrogen decline further predisposes women to LV diastolic dysfunction by means of renin-angiotensin-aldosterone system activation and altered kinase-mediated titin phosphorylation (26). These mechanistic distinctions support sex-stratified screening strategies. As noted in reviews from the past decade, LA function captures early diastolic stress and "prefailure" signals often missed by ventricular indexes alone (27). In men with predominant structural remodeling, the

predictive advantage of LA ejection fraction supports its routine clinical reporting. Conversely, in women, HF most commonly manifests as HF with preserved ejection fraction (25). Although LVEF remained a key predictor, the current non-sex-specific 50% ejection fraction threshold may introduce diagnostic bias (25). Screening in women should not rely solely on apparently normal LVEF but instead prioritize myocardial strain imaging to detect subclinical dysfunction from microvascular inflammation, addressing gaps in conventional guidelines (28).

Both circumferential and radial myocardial strains were associated with higher HF risk in men and women, providing more stable predictive performance than cardiac output or large-vessel parameters (29). Circumferential strain reflects midwall myocardial fiber function and serves as a sensitive marker of subclinical LV dysfunction, with high incremental predictive value (30). In

Table 4: Comparison of Predictive Performance of the PREVENT + Cardiac MRI Model and Existing Clinical HF Risk Scores

Subgroup and Model	Training Set		Internal Test Set	
	C-index	P Value*	C-index	P Value*
Overall†				
PREVENT + cardiac MRI	0.824 (0.746, 0.902)	Ref	0.847 (0.790, 0.904)	Ref
ARIC	0.712 (0.625, 0.799)	.004	0.776 (0.705, 0.847)	.09
PCP-HF	0.712 (0.625, 0.798)	.001	0.806 (0.737, 0.875)	.34
Men‡				
PREVENT + cardiac MRI	0.866 (0.746, 0.986)	Ref	0.756 (0.669, 0.844)	Ref
ARIC	0.689 (0.532, 0.845)	.015	0.728 (0.636, 0.821)	.63
PCP-HF	0.716 (0.547, 0.885)	.042	0.752 (0.657, 0.846)	.93
Women§				
PREVENT + cardiac MRI	0.843 (0.760, 0.926)	Ref	0.861 (0.815, 0.900)	Ref
ARIC	0.754 (0.654, 0.855)	.039	0.804 (0.718, 0.890)	<.001
PCP-HF	0.789 (0.692, 0.885)	.18	0.793 (0.688, 0.899)	.005

Note.—Data in parentheses are 95% CIs. This analysis was restricted to participants with complete data for all variables required to calculate the external risk scores (Atherosclerosis Risk in Communities heart failure [HF] risk score [ARIC] and Pooled Cohort Equations to Prevent Heart Failure [PCP-HF]). PREVENT = Predicting Risk of Cardiovascular Disease EVENTS, Ref = reference.

* P values were calculated using the DeLong test to compare the C-index of each comparator model (ARIC or PCP-HF) against the PREVENT + cardiac MRI model (reference).

† Overall: training set, number of events = 150 of 18 249 participants (0.82%); internal test set, number of events = 62 of 7618 (0.81%).

‡ Men: training set, number of events = 93 of 8585 (1.08%); internal test set, number of events = 46 of 3628 (1.27%).

§ Women: training set, number of events = 57 of 9664 (0.59%); internal test set, number of events = 16 of 3990 (0.40%).

our study, it also demonstrated incremental predictive value (Δ C-index = 0.008 in women and 0.009 in men).

Aortic cardiac MRI indexes, although not the strongest predictors, offer complementary pathophysiologic insights. Measures such as minimal aortic area and distensibility reflect vascular stiffness and early ventricular-arterial uncoupling. Reduced aortic compliance increases LV systolic load and impairs diastolic perfusion, promoting ischemia, fibrosis, and remodeling, accelerating HF with preserved ejection fraction onset (31). This aligns with Multi-Ethnic Study of Atherosclerosis findings, where ascending aortic distensibility independently predicted mortality and cardiovascular events, highlighting its added value beyond traditional HF risk parameters (32).

Although echocardiography remains the primary screening tool, its sensitivity is limited by acoustic windows and a lack of tissue characterization. Cardiac MRI provides superior reproducibility and unique tissue phenotyping, particularly valuable in patients with poor acoustic windows (33). Rapid scanning protocols (<10 minutes) and artificial intelligence–automated analysis are reducing costs and interpretation burdens, and health economic analyses indicate that cardiac MRI reduces downstream expenditures by minimizing unnecessary testing (33,34). As these technologies mature, cardiac MRI–based risk stratification will become increasingly cost-effective for broader population screening.

This study has several limitations. First, the lack of a large-scale independent cohort with cardiac MRI data prevented external validation of model generalizability; accordingly, this model should not be applied in clinical practice until rigorously evaluated in an independent external cohort. Furthermore, UK Biobank participants were predominantly White, and HF risk profiles may vary across geographic regions. Direct application to other populations may cause biased risk stratification and

worsen health inequalities; validation in geographically and ethnically diverse cohorts is needed before extrapolation to other ancestry groups. Second, due to limited incident cases (training set: HF with reduced ejection fraction, $n = 31$ and HF with preserved ejection fraction, $n = 224$; internal test set: HF with reduced ejection fraction, $n = 15$ and HF with preserved ejection fraction, $n = 97$), we could not perform HF subtype–specific analyses. Larger studies are needed to investigate the predictive value of cardiac MRI parameters across specific HF subtypes. Third, the limited number of events in women ($n = 37$) poses an overfitting risk; however, a high shrinkage factor (0.9682) and consistent predictor selection across stepwise and random forest methods support model robustness, although cautious interpretation is warranted pending evaluation in larger samples. Finally, voluntary enrollment of healthy participants may introduce selection bias; multivariable models should be validated in larger, multicenter populations.

In conclusion, integrating cardiac MRI parameters into the Predicting Risk of Cardiovascular Disease EVENTS equations not only improved heart failure risk prediction but also revealed sex-specific pathophysiologic differences, underscoring the predictive value of multidimensional imaging parameters beyond conventional risk factors, particularly in sex-stratified risk assessment. Future studies should replicate these findings in large, multicenter, and ethnically diverse cohorts to further enhance the clinical translation of this model.

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